TECHNICAL DOCUMENTATION | ELENA GEO TECH EVK V3 | NAVIC & GNSS EVALUATOR V1.0

Elena Geo Tech EVK v3 — GNSS & NavIC Performance Evaluator

Comprehensive Technical Specification, Geodetic Formulations, Architecture Manual & System Blueprint

HARDWARE KIT	CONSTELLATIONS	DOCUMENT TYPE	DATE
Elena Geo Tech EVK v3	NavIC (IRNSS), GPS, GLONASS	Engineering System Manual	November 2026

1. About the Software

1.1 Executive Overview & Purpose

The **Elena Geo Tech EVK v3 GNSS & NavIC Performance Evaluator** is a dedicated, desktop-based scientific evaluation software designed to audit, benchmark, and visualize the performance of multi-constellation satellite navigation receivers. Developed specifically for testing the **Elena Geo Tech EVK v3 evaluation kit and high-gain antenna**, the software evaluates real-world raw NMEA 0183 data extracted from field loggers.

The software bridges the gap between raw hardware logging and high-level client deliverables by providing:

- **Decoupled Standalone Operation:** Accepts any NMEA `.txt` file generated by the logger hardware. No physical hardware connection or serial driver locking is required on the evaluation workstation.
- **Automated Pre-Ingestion Heuristic Validation:** Conducts an instant (sub-20 ms) sanity check on the uploaded file to ensure valid NMEA 0183 checksums and structure before running complex computations.
- **The 4 International Benchmark Pillars:** Automatically computes Local ENU position dispersion, RF signal quality & multipath attenuation, constellation geometry (DOP), and fix reliability.
- **Publication-Grade Engineering Reporting:** Generates auditable, self-contained HTML5 and PDF technical reports suitable for defense clients, ISRO reviews, and academic viva presentations.

1.2 The Problem Solved

Field receivers produce continuous ASCII serial streams containing tens of thousands of fragmented, asynchronous NMEA sentences (`$GGA` , `$RMC` , `$GSA` , `$GSV`). Standard spreadsheets cannot properly analyze this data because:

1. Positions are in geodetic curvilinear coordinates (Latitude/Longitude/Altitude on the WGS-84 ellipsoid) and must be rotated into an orthogonal topocentric Cartesian frame (Local East, North, Up) to assess sub-meter radial accuracy.
2. RF signal strengths (C/N_0) arrive split across multiple satellite vehicle messages and must be correlated with spatial elevation angles to compute low-elevation multipath degradation.
3. Constellation talker IDs for India's indigenous NavIC constellation (`$GI` / `$IN`) differ from standard GPS (`$GP`), requiring multi-talker epoch synchronization.

2. What's Inside the Software (Architecture & Inventory)

2.1 Directory Structure

```
gnss_navic_logger/
│
├── evaluator_app.py           # Primary Tkinter Desktop Application (GUI, Async Threading, Canvas)
├── main.py                    # CLI & GUI entry point (supports automated batch evaluations)
├── README.md                  # Quickstart documentation & PyCharm run guide
├── requirements.txt           # Python dependencies (numpy, matplotlib)
│
├── core/                      # Core Mathematical & Analytical Engines
│   ├── evaluator.py           # Pillar calculations, geodetic math, pre-validation scanner
│   ├── nmea_parser.py         # NMEA 0183 sentence decoder, talker ID resolver, epoch synchronizer
│   ├── plotter.py             # In-place Matplotlib figure renderer for the 4 pillars
│   └── report_generator.py    # Standalone HTML5/PDF publication report generator
│
├── data/                      # Local Data Store
│   ├── samples/               # Sample NMEA .txt files (standard, zero-fix, single-fix)
│   ├── reports/               # Automatically generated HTML/PDF engineering reports
│   └── plots/                 # Exported companion PNG plots (ENU scatter, CDF, skyplot, RF)
│
└── test_app_upload.py         # Automated regression & stability verification test suite
```

2.2 Component & Module Breakdown

Module File	Primary Class / Components	Functional Role
`evaluator_app.py`	`StandaloneEvaluatorApp`	Tkinter GUI container, dark engineering theme, asynchronous worker thread dispatcher, 4 headline KPI cards, 4 interactive tabs with Matplotlib canvas integration.
`core/evaluator.py`	`GNSSEvaluator`	Implements all mathematical algorithms for the 4 pillars, coordinate frame rotations (WGS-84 to ECEF to ENU), statistical dispersion estimators, and file pre-validation.
`core/nmea_parser.py`	`NMEAParser`, `NMEAFix`	Validates 8-bit XOR checksums, parses geographical coordinates from DDMM.MMMM to decimal degrees, resolves multi-constellation talker IDs, and buffers epoch records.
`core/plotter.py`	`GNSSPlotter`	Generates horizontal ENU scatter, Error CDF curves, C/N0 vs Elevation angle multipath plots, and polar stereographic skyplots with in-place figure recycling.
`core/report_generator.py`	`ReportGenerator`	Assembles an executive HTML5/CSS3 report containing summary KPI badges, geodetic coordinate tables, satellite PRN tracking matrices, and print-ready formatting.

2.3 End-to-End Processing Pipeline

Dataflow Pipeline:
1. Upload (User picks `.txt` file) → **2. Validation** (Fast 20 ms heuristic scanner verifies NMEA sentences) → **3. Worker Thread** (Background ingestion streams lines without blocking UI) → **4. Multi-Sentence Epoch Fusion** (Fuses `$GGA`, `$RMC`, `$GSA`, `$GSV` into synchronized epochs) → **5. Geodetic & Statistical Computation** (Local ENU, DRMS, CEP, C/N0 separation, DOP analysis) → **6. UI Refresh & Reporting** (Thread-safe `root.after()` callback updates cards, redraws Matplotlib figures in-place, and unlocks report export).

3. Mathematical Formulations, Algorithms & Methods

The evaluation engine implements rigorous geodetic transformations and statistical estimators adhering to international GNSS benchmarking standards (e.g., RTCM, US DoD GPS SPS, and ISRO IRNSS performance standards).

3.1 Pillar 1: Positioning Accuracy & Geodetic Dispersion

1. Geodetic to ECEF (Earth-Centered, Earth-Fixed) Transformation

Receiver positions are converted from geodetic latitude ϕ, longitude λ, and ellipsoidal height h to Cartesian ECEF coordinates (X, Y, Z) using the **WGS-84 reference ellipsoid**:

$$N(\phi) = \frac{a}{\sqrt{1 - e^2 \sin^2(\phi)}}$$

$$X = (N(\phi) + h) \cos(\phi) \cos(\lambda)$$

$$Y = (N(\phi) + h) \cos(\phi) \sin(\lambda)$$

$$Z = \left(N(\phi)(1 - e^2) + h\right) \sin(\phi)$$

Where:
• Semi-major axis $a = 6,378,137.0 \text{ meters}$
• Flattening $f = 1/298.257223563$
• First eccentricity squared $e^2 = 2f - f^2 \approx 0.00669437999014$

2. ECEF to Local Tangent Plane (Local ENU: East, North, Up)

To evaluate horizontal and vertical positioning errors independent of global curvature, the ECEF coordinate displacements are rotated into a topocentric Local Tangent Plane centered at the antenna reference position (ϕ_0, λ_0, h_0):

$$\begin{bmatrix} \Delta E \\ \Delta N \\ \Delta U \end{bmatrix} = \begin{bmatrix} -\sin(\lambda_0) & \cos(\lambda_0) & 0 \\ -\sin(\phi_0)\cos(\lambda_0) & -\sin(\phi_0)\sin(\lambda_0) & \cos(\phi_0) \\ \cos(\phi_0)\cos(\lambda_0) & \cos(\phi_0)\sin(\lambda_0) & \sin(\phi_0) \end{bmatrix} \begin{bmatrix} X - X_0 \\ Y - Y_0 \\ Z - Z_0 \end{bmatrix}$$

If the user provides an external ground-truth surveyed coordinate, that coordinate is used as (ϕ_0, λ_0, h_0) to compute *absolute accuracy*. If no reference is provided, the mathematical centroid mean of the session is used to evaluate *relative repeatable dispersion*.

3. Statistical Dispersion & Error Figures of Merit

- **Component Standard Deviations:**

$$\sigma_E = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\Delta E_i - \bar{E})^2}, \quad \sigma_N = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\Delta N_i - \bar{N})^2}, \quad \sigma_U = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\Delta U_i - \bar{U})^2}$$

- **2D Horizontal Radial Error (r_{2D}) & 3D Spherical Error (r_{3D}):**

$$r_{2D,i} = \sqrt{\Delta E_i^2 + \Delta N_i^2}, \qquad r_{3D,i} = \sqrt{\Delta E_i^2 + \Delta N_i^2 + \Delta U_i^2}$$

- **DRMS (Distance Root Mean Square):**

$$\mathrm{DRMS} = \sqrt{\sigma_E^2 + \sigma_N^2} \quad (\approx 63.2\% - 68.3\% \text{ probability})$$

- **2DRMS (Twice Distance Root Mean Square — International Benchmark Standard):**

$$2\mathrm{DRMS} = 2 \times \mathrm{DRMS} = 2\sqrt{\sigma_E^2 + \sigma_N^2} \quad (\approx 95.4\% - 98.2\% \text{ probability})$$

- **CEP (Circular Error Probable — 50% Median Radius):**

$$\mathrm{CEP}_{50} = \mathrm{Percentile}_{50}(r_{2D})$$

- **SEP (Spherical Error Probable — 3D 50% Sphere):**

$$\mathrm{SEP} = \mathrm{Percentile}_{50}(r_{3D})$$

- **Vertical Dispersion ($2\sigma_U$):**

$$2\sigma_U = 2 \times \sigma_U \quad (\approx 95.4\% \text{ vertical confidence interval})$$

- **Empirical Error CDF (Cumulative Distribution Function):** The radial errors r_{2D} are sorted in ascending order $r_{(1)} \leq r_{(2)} \leq \cdots \leq r_{(N)}$ with assigned cumulative probabilities $P(r) = \frac{i}{N} \times 100\%$. The evaluator calculates the 50th (CEP), 68.27th (1σ), 95th ($2\mathrm{DRMS}$), and 99th percentiles.

3.2 Pillar 2: RF Signal Quality & Tracking Attributes

Carrier-to-Noise Density Ratio (C/N_0)

C/N_0 indicates the received satellite signal power relative to thermal noise floor, extracted from NMEA `$GSV` sentences in units of $\mathrm{dB\text{-}Hz}$.

$$\overline{C/N_0} = \frac{1}{K}\sum_{k=1}^{K}(C/N_0)_k$$

- **Frequency Band & Constellation Separation:** The software dynamically catalogs satellites by constellation and band:
  - **NavIC (IRNSS):** Tracked on L5 (1176.45 MHz) and S-Band (2492.028 MHz), PRNs 01–14.
  - **GPS:** Tracked on L1 (1575.42 MHz) and L5, PRNs 01–32.
  - **GLONASS:** Tracked on G1/G2 FDMA, PRNs 65–96.
- **Elevation-Dependent Multipath Modeling:** Low-elevation signals travel through denser troposphere layers and reflect off nearby terrain/structures. The evaluator applies an international standard 15° **elevation cut-off mask**:

$$\text{Multipath Attenuation } (\Delta\text{dB}) = \overline{C/N_0}_{(\text{Elevation}\geq 15^\circ)} - \overline{C/N_0}_{(\text{Elevation}<15^\circ)}$$

  A low $\Delta\mathrm{dB}$ (≤ 4 dB) indicates superior multipath rejection by the receiver antenna.
- **Satellite Tracking & Usage Efficiency:**

$$\text{Usage Efficiency } (\%) = \frac{\text{Mean Satellites Used } (\overline{N}_{\text{used}})}{\text{Mean Satellites in View } (\overline{N}_{\text{view}})} \times 100\%$$

3.3 Pillar 3: Satellite Geometry & Constellation Health

Dilution of Precision (DOP) measures the geometric configuration of satellites in view. Poor geometry clusters satellites in one sector of the sky, magnifying range errors into large positioning uncertainties:

$$\text{Positioning Error} \approx \text{DOP} \times \sigma_{\text{UERE}}$$

DOP Metric	Mathematical Relationship	Target Health Thresholds
HDOP (Horizontal DOP)	$\sigma_H = \text{HDOP} \cdot \sigma_0$	≤ 1.0 (Ideal), $1.0 - 2.0$ (Excellent)
VDOP (Vertical DOP)	$\sigma_V = \text{VDOP} \cdot \sigma_0$	≤ 1.5 (Ideal), $1.5 - 2.5$ (Good)
PDOP (3D Position DOP)	$\text{PDOP} = \sqrt{\text{HDOP}^2 + \text{VDOP}^2}$	• **< 2.0:** Optimal Geometry • **2.0 – 3.5:** Good Geometry • **> 3.5:** Degraded Geometry

Polar Stereographic Skyplot Projection

Satellites with azimuth $\alpha \in [0^\circ, 360^\circ]$ and elevation $\theta \in [0^\circ, 90^\circ]$ are mapped onto polar coordinates:

$$r = 90^\circ - \theta, \qquad \theta_{\text{polar}} = \alpha \quad (\text{North at } 0^\circ,\ \text{clockwise})$$

This plot reveals the distinctive stationary GEO/GSO footprint of NavIC over the Indian subcontinent alongside dynamic GPS orbital passes.

3.4 Pillar 4: Fix Reliability & Temporal Continuity

- **Fix Availability Ratio:**

$$\text{Availability } (\%) = \frac{\text{Total Valid 3D Fix Epochs } (GGA \text{ Quality} \geq 1)}{\text{Total Logged Epochs}} \times 100\%$$

- **NMEA Fix Quality Classifications:** Extracted from Field 6 of `$GGA`:
  - `0`: Invalid / No Position Fix
  - `1`: Autonomous SPS (Standard Positioning Service)
  - `2`: Differential GPS (DGPS / SBAS)
  - `4`: RTK Fixed (Centimeter Carrier-Phase Solution)
  - `5`: RTK Float
- **TTFF (Time to First Fix):**

$$\text{TTFF} = t_{\text{first valid 3D fix}} - t_{\text{receiver power-on / log start}}$$

- **Epoch Continuity & Rate Jitter:** For successive timestamps t_i and t_{i-1}, the epoch delta is $\Delta t_i = t_i - t_{i-1}$:

$$\bar{\Delta t} = \frac{1}{N-1}\sum_{i=2}^{N} \Delta t_i, \qquad \text{Nominal Rate (Hz)} = \frac{1}{\bar{\Delta t}}, \qquad \text{Jitter} = \sigma_{\Delta t}$$

  If $\Delta t_i > 1.8 \text{ seconds}$ (for a standard 1 Hz logger), missing epochs are flagged as **Dropped Frames**:

$$\text{Dropped Frames} = \sum \left(\text{round}(\Delta t_i) - 1\right)$$

---

4. Technical Stack & Development Tools

Technology / Library	Version / Specs	Role & Justification
Python	3.10 – 3.14 (64-bit)	Core language runtime; provides geodetic arithmetic, file IO, and threading primitives.
Tkinter & ttk	Standard Library (Tcl/Tk 8.6)	Desktop GUI toolkit; delivers a native Windows window with zero heavy web/Electron overhead.
NumPy (`numpy`)	≥ 1.24.0	Vectorized matrix math, standard deviations, array sorting, and empirical percentiles.
Matplotlib (`matplotlib`)	≥ 3.7.0 (`Agg` backend)	High-resolution scientific plotting; embedded into Tkinter via `FigureCanvasTkAgg`.
Python `threading`	Standard Library	Executes NMEA stream reading and heavy math in a background daemon thread, keeping the GUI at 60 FPS.
HTML5 & CSS3	Modern Spec	Generates standalone, CSS-styled publication reports with `@media print` vector PDF rendering.
KaTeX / MathJax	0.16.8	Typesets mathematical formulas and Greek symbols into clean vector typography for PDF generation.

5. Architectural Safeguards & Reliability Engineering

> **Key Reliability Features Built Into the Architecture:**

1. **Instant Pre-Ingestion Heuristic Validation:** Before allocating large memory structures, the software samples the first 5,000 lines of an uploaded file. It verifies NMEA 0183 talker headers and displays a confirmation dialog showing line counts, sentence types, and constellations found. If an invalid text file is uploaded, it safely rejects it without crashing.
2. **In-Place Canvas Figure Recycling:** Previous revisions experienced Windows Tcl/Tk crashes when calling `plt.close()` and reassigning canvas widget figures. The plotter was re-engineered to clear existing figures in-place (`fig.clf()`) and trigger non-blocking flushes via `canvas.draw_idle()`, guaranteeing zero handle leaks.
3. **Daemon Thread UI Decoupling:** Large NMEA files (10,000+ lines) are parsed and evaluated inside a background worker thread (`threading.Thread(daemon=True)`). The main thread remains responsive, eliminating the Windows "(Not Responding)" lockup.
4. **Two-Tier Responsive Headline Cards:** Dashboard KPI cards use a two-line layout (large bold primary metric on top, secondary details below). This prevents text overflow or horizontal clipping on high-DPI displays.
5. **Zero-Fix & Cold-Start Immunity:** Field logs recorded indoors or during receiver cold starts often contain satellite `$GSV` messages but zero valid 3D coordinates. The evaluation engine populates default structures for positioning accuracy rather than throwing `KeyError`, allowing the RF and geometry tabs to still display satellite data.

---

6. PyCharm Run & Deployment Guide

6.1 Running in PyCharm

1. Open the project directory in **PyCharm**:
   `C:\Users\Gowtham c\.gemini\antigravity\scratch\gnss_navic_logger`
2. In the top-right toolbar next to the green Play button, select **Run Evaluator App** and click **Play ▶**.
3. Alternatively, open the PyCharm terminal and execute:
   `py evaluator_app.py`
4. Click **" 📤 Upload NMEA Data File (.txt)"** and select your extracted logger file.
5. Review the validation popup, click **OK**, and inspect the 4 benchmark tabs.
6. Click **" 📄 Export PDF / HTML Report"** to generate the client-ready evaluation deliverable.

6.2 Command-Line Automated Mode (Headless / Batch)

```
# Evaluate a single file via CLI
py main.py --evaluate "data/samples/sample_logger_output.txt"

# Evaluate with known ground-truth benchmark coordinates
py main.py --evaluate "data/samples/sample_logger_output.txt" --ref-lat 12.9716 --ref-lon 77.5946 --ref-alt 920.0

# Run automated self-test on simulated Elena EVK v3 log
py main.py --test
```

> **Elena Geo Tech EVK v3 Performance Evaluator**
> Engineered for High-Reliability GNSS & NavIC Verification • Built with Python, Tkinter & NumPy